BAYMAX

A locally-runnable personal-task agent with an eval harness. Stepping-stone learning project — shipped as portfolio evidence of engineering + methodology discipline, not as a daily-driver assistant or research contribution.

Status: Closed out by the project owner on 2026-10-17 as a stepping-stone learning project. The final local fine-tune records 73/150 task successes on the development suite. Fake adapters only; not a production assistant.

See docs/PROJECT.md for the full "what BAYMAX is and isn't."

What it does

1. Accepts text input describing a productivity task
2. Selects and executes tool calls against **fake** in-memory adapters (calendar / notion / gmail / clipboard)
3. Confirms, refuses, or requests clarification
4. Logs actions with trace / task identifiers
5. Runs against a scripted eval suite; produces metrics JSON

What it doesn't do (honest scope)

- **No real API integration** — fake adapters only. Not connected to your real Google Calendar / Gmail / Notion.
- **Not a daily-driver assistant** — response quality too low to trust for real actions
- **No menu bar app / voice / chat UI** — REST endpoint only (`POST /invoke`)
- **No context memory across requests** — each request stateless
- **No streaming, no multi-turn dialog**

Full list in docs/DEFERRED.md.

Team

Name	Owned modules
Marv	Agent core, inference service, backends (OpenAI + MLX), output adapter, telemetry
Ronin	Eval harness, training pipeline, tool adapters (fake), scenario authorship

Full ownership rules: docs/OWNERSHIP.md.

Documentation

Doc	Purpose
docs/PROJECT.md	What BAYMAX is (and isn't) — the framing doc
docs/ARCHITECTURE.md	System design
docs/OWNERSHIP.md	Module ownership + PR rules
docs/ROADMAP.md	Timeline to ship (~5 weeks from 2026-08-24)
docs/BACKEND_FLOW.md	Backend + agent design diagrams
docs/BENCHMARKING.md	Eval methodology + results
docs/SCORING.md	Deterministic scoring rules
docs/DEFERRED.md	Known limitations + not-implemented items
docs/decisions/	ADR archive
docs/experiments/README.md	Experiment history, settings, results and limitations
Experiment notebook PDF	Combined experiment reports with internal navigation
Complete documentation PDF	Public project documents and experiment reports

Quick start

Install dependencies:

```
uv sync --frozen --all-groups
```

Run checks:

```
uv run ruff format --check
uv run ruff check .
uv run pyright
uv run pytest
```

Run the scripted baseline:

```
python -m baymax.eval.cli run-scripted \
  --scenarios scenarios/v1 \
  --responses scenarios/v1/scripted_responses/v1-scripted.json \
  --output results/phase1/scripted/v1-baseline.json
```

Run the OpenAI agent baseline:

```
export OPENAI_API_KEY="your_api_key"
python -m baymax.eval.cli run-agent-openai \
  --scenarios scenarios/v1 \
  --output results/phase1/openai/gpt-4o-mini-full.json \
  --model gpt-4o-mini
```

Phase 1 result files are committed under `results/`.

Final local evaluation (150 development scenarios)

Local checkpoint	Task successes / 150	Routing macro-F1	Invalid outputs / 150
Unweighted name-first reference, 1k / 2 epochs	57	0.612	1
Corrected grounding, 1k / 2 epochs	53	0.591	0
Expanded coverage, 2k / 2 epochs	64	0.615	2
Expanded coverage, 2k / 4 epochs	73	0.681	0

Source: saved four-epoch evaluation. All four use compact JSON, greedy decoding and the same combined v1/v2 suite. This is single-seed development evidence after repeated evaluation, not an untouched test set or proof of statistical significance. The final run correctly routes 17/40 clarification cases and 27/29 refusal cases; clarification remains a weakness. Its recorded mean latency is 5,652 ms, versus 2,758 ms for the two-epoch 2k run. Hardware conditions were not controlled sufficiently to attribute that difference to epochs alone. Reported API cost is zero, not zero hardware/electricity cost.

Recipes, decision matrices and qualifications are in the four-epoch experiment report. Generated training data, adapter weights and private local files are not included in Git. The committed config pins the dataset revision; reproducing or serving the local model requires generating the data and obtaining/training its adapter.

Historical Phase 1 baselines (50 scenarios)

Backend	task_success_rate	scenarios
Scripted baseline	see `results/phase1/scripted/v1-baseline.json`	50
gpt-4o-mini	**0.52**	50
Qwen 1.5B + LoRA (smoke test)	0.24	50

These historical 50-case results are not directly comparable to the final 150-case local evaluations above. No claim that the fine-tune beats a frontier model is made.

License

MIT — Marv and Ronin, 2026.

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